

Experiment No. 1 — Linear Regression (Housing Price Prediction)

Linear Regression is one of the simplest and most important Machine Learning algorithms used for predicting **continuous values** like:

- House prices
- Salary prediction
- Temperature forecasting
- Stock trends

In this experiment, you will:

1. Understand the concept
2. Learn the mathematical idea
3. Implement it in Python
4. Train and test the model
5. Evaluate accuracy
6. Predict house prices

1. What is Linear Regression?

Linear Regression finds the relationship between:

- **Independent variables (X)** → Features
- **Dependent variable (Y)** → Target

Example:

- Area of house
- Number of bedrooms
- Age of house

These features help predict:

- House price

2. Real-Life Example

Suppose:

Area (sq.ft)	Price
1000	25 Lakhs
1500	40 Lakhs
2000	55 Lakhs

As area increases, price also increases.

Linear regression tries to fit a straight line:

$$y = mx + c$$

Where:

- y = predicted output
- x = input feature
- m = slope
- c = intercept

3. Types of Linear Regression

A) Simple Linear Regression

One input variable.

Example:

- Area $\rightarrow$ Price

B) Multiple Linear Regression

Multiple input variables.

Example:

- Area
- Bedrooms
- Bathrooms
- Age

→ Predict Price

4. Workflow of the Experiment

```
Dataset Collection
    ↓
Data Preprocessing
    ↓
Train-Test Split
    ↓
Model Training
    ↓
Prediction
    ↓
Evaluation
```

5. Required Libraries

Install if needed:

```
pip install pandas numpy matplotlib scikit-learn
```

6. Complete Python Implementation

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

7. Create Sample Housing Dataset

```
data = {
    'Area': [1000, 1500, 1800, 2400, 3000, 3500, 4000],
    'Bedrooms': [2, 3, 3, 4, 4, 5, 5],
    'Price': [25, 40, 45, 65, 75, 85, 95]
}
```

```
df = pd.DataFrame(data)

print(df)
```

8. Define Features and Target

```
X = df[['Area', 'Bedrooms']] # Predictor variables
y = df['Price']               # Target variable
```

9. Split Dataset

Training data → Learn
Testing data → Evaluate

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

10. Train the Linear Regression Model

```
model = LinearRegression()

model.fit(X_train, y_train)
```

11. Predict House Prices

```
y_pred = model.predict(X_test)

print("Predicted Prices:")
print(y_pred)
```

12. Evaluate the Model

Mean Squared Error (MSE)

Lower MSE = Better model

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

```
mse = mean_squared_error(y_test, y_pred)

print("Mean Squared Error:", mse)
```

R² Score

Measures model accuracy.

- Closer to 1 → Better model
- Closer to 0 → Poor model

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

```
r2 = r2_score(y_test, y_pred)

print("R2 Score:", r2)
```

13. Visualize the Regression Line

For simple visualization using Area only:

```
plt.scatter(df['Area'], df['Price'], color='blue')

plt.plot(df['Area'],
         model.predict(df[['Area', 'Bedrooms']]),
         color='red')

plt.xlabel("Area")
plt.ylabel("Price")
plt.title("Linear Regression - Housing Prices")

plt.show()
```

14. Predict New House Price

Example:

- Area = 2500
- Bedrooms = 4

```
new_house = [[2500, 4]]

predicted_price = model.predict(new_house)

print("Predicted House Price:", predicted_price[0], "Lakhs")
```

15. Expected Output

Predicted Prices:
[64.2 42.5]

Mean Squared Error: 3.12

R2 Score: 0.95

(Values may vary slightly.)

16. Viva Questions with Answers

Q1. What is Linear Regression?

Linear Regression is a supervised machine learning algorithm used to predict continuous values.

Q2. What is the difference between dependent and independent variables?

- Independent Variable → Input features
 - Dependent Variable → Output/Target
-

Q3. What is the equation of linear regression?

$$y = mx + c$$

Q4. What is the use of train-test split?

It separates training data and testing data for proper evaluation.

Q5. What is R² score?

It measures how well the model fits the data.

Q6. What is overfitting?

When model performs well on training data but poorly on unseen data.

17. Short Conclusion

Thus, the Linear Regression model was successfully implemented to predict housing prices based on house features. The model was trained, tested, and evaluated using performance metrics like MSE and R² score.

18. Full Final Code (Ready to Run)

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Dataset
data = {
    'Area': [1000, 1500, 1800, 2400, 3000, 3500, 4000],
    'Bedrooms': [2, 3, 3, 4, 4, 5, 5],
    'Price': [25, 40, 45, 65, 75, 85, 95]
}

df = pd.DataFrame(data)

# Features and Target
X = df[['Area', 'Bedrooms']]
y = df['Price']

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Model
```

```
model = LinearRegression()

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluation
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Predicted Values:", y_pred)
print("MSE:", mse)
print("R2 Score:", r2)

# Predict new data
new_house = [[2500, 4]]

predicted_price = model.predict(new_house)

print("Predicted House Price:",
      predicted_price[0], "Lakhs")

# Visualization
plt.scatter(df['Area'], df['Price'], color='blue')

plt.plot(df['Area'],
         model.predict(df[['Area', 'Bedrooms']]),
         color='red')

plt.xlabel("Area")
plt.ylabel("Price")
plt.title("Linear Regression")

plt.show()
```